

Time-Aware Inertial Normalization for Irregularly-Sampled Tabular Streams

Tuhan Agay

Segmora AI · London, United Kingdom

research@segmora.ai · segmora.ai

arXiv Preprint · 2026

Abstract

Standard normalization layers in neural networks are time-blind: the Exponential Moving Average (EMA) that maintains running statistics applies a fixed forgetting weight α regardless of the elapsed time between observations.

We introduce **Time-Aware Inertial Normalization (TAIN)**, which replaces the fixed EMA coefficient with $\alpha^{\Delta t}$, where Δt is the real time gap between consecutive observations. The formula $\alpha^{\Delta t} = e^{-\lambda \Delta t}$ is well established for exponential smoothing of irregular time series [Wright, 1986, Zumbach & Müller, 2001]; our contribution is applying it to the *normalization layer’s running statistics*, a component that all prior irregular time series methods (GRU-D, Neural ODEs, mTAN) leave untouched. We show that $\alpha^{\Delta t}$ is the natural discretization of the Ornstein-Uhlenbeck process, grounding TAIN in continuous-time stochastic process theory.

To evaluate TAIN’s impact on action stability, we analyze jitter using a decomposition into *signal jitter* (desirable tracking of genuine signal movement) and *unnecessary jitter* (undesirable oscillation exceeding signal change), combined with flip rate and directional agreement metrics drawn from signal processing and econometrics. No prior work on policy churn [Schaul et al., 2022] or action oscillation [Chen et al., 2021] provides this decomposition.

Empirical validation on five real-world datasets (Retail, Sensor, Finance, ICU-Temp, ICU-Urine; 5,409 entities, 659,325 observations) demonstrates that TAIN achieves statistically significant RMSE improvements over standard EMA ($p < 0.001$ in four of five domains) while maintaining the lowest unnecessary jitter ratio among RMSE-improving methods. Post-gap recovery scales monotonically with gap magnitude, directly confirming the Ornstein-Uhlenbeck theoretical prediction.

Code is available at <https://github.com/segmora/TAIN>.

1. Introduction

Machine learning on tabular data has long been dominated by two families of models: gradient-boosted decision trees (GBDT) and multilayer perceptrons (MLP). Both share a fundamental assumption: observations arrive independently, at uniform intervals, with the sole objective of minimizing a pointwise prediction loss. This assumption fails systematically in operational AI settings.

Consider a retail chain deploying a store management intelligence system. Daily sales figures arrive on weekdays, but weekend data is aggregated; public holiday gaps of 7–14 days interrupt

otherwise regular streams; sensor outages introduce sporadic missingness. The temporal structure of the data (the fact that a 30-day gap carries different information than a 1-day gap) is invisible to any model that treats each record as an i.i.d. sample.

The consequence goes beyond suboptimal forecasting. The downstream objective is not prediction but *action*: a store manager receives a daily directive (reorder this SKU, redeploy staff, trigger a promotion). When the normalization layer is time-blind, two observations separated by a holiday gap are treated identically to two observations separated by a single day. The model’s internal statistics carry stale momentum; the generated actions oscillate. We call this **action jitter**, and it is the primary cause of trust erosion in operational AI deployments.

This paper addresses two gaps in the literature. First, while methods such as GRU-D [Chen et al., 2018], Neural ODEs [Chen et al., 2018], and mTAN [Shukla & Marlin, 2021] introduce time-awareness into hidden state dynamics or attention mechanisms, *none modifies the normalization layer*. The running statistics that BatchNorm maintains via EMA are updated with a fixed α regardless of observation spacing. Second, while policy churn [Schaul et al., 2022] and action oscillation [Chen et al., 2021] have been identified as problems, no standard methodology decomposes jitter into its desirable (signal-tracking) and undesirable (noise-driven) components.

1.1 Contributions

- **TAIN**: a normalization layer in which the inertia coefficient $\alpha^{\Delta t}$ adapts to the real elapsed time between observations, derived as the natural discretization of the Ornstein-Uhlenbeck process. The formula $\alpha^{\Delta t}$ is well known in time series analysis [Wright, 1986, Zumbach & Müller, 2001]; our contribution is its application to the normalization layer, a component untouched by prior irregular time series methods.
- **Jitter decomposition analysis**: we evaluate TAIN using a decomposition of total estimate jitter into signal jitter (desirable) and unnecessary jitter (undesirable), complemented by flip rate and directional agreement, all drawn from signal processing and econometrics. This analytical lens enables principled evaluation of the RMSE–jitter trade-off.
- **Empirical validation** on five real-world datasets (5,409 entities, 659,325 observations) against six baselines including the Kalman Filter, Holt Exponential Smoothing, and Double EMA. TAIN is shown to achieve the best RMSE and post-gap recovery within the low-jitter operating regime.

2. Background and Related Work

2.1 Neural ODEs and Irregular Time Series

Neural ODEs [Chen et al., 2018] define the hidden state $\mathbf{h}(t)$ as a continuous-time quantity governed by a learned vector field:

$$\frac{d}{dt}\mathbf{h}(t) = f(\mathbf{h}(t), \mathbf{x}(t), \theta), \quad (1)$$

where $\mathbf{x}(t)$ is the input at time t and θ are learnable parameters. The continuous-time formulation handles irregular observation times naturally. Latent ODEs [Rubanova et al., 2019] extend this to sequences with structured missingness. Neural CDEs [Kidger et al., 2020] extend the formulation to controlled dynamics driven by the data path. Liquid Time-Constant Networks [Hasani et al., 2021] introduce input-dependent time constants into ODE-based recurrent

architectures. Multi-Time Attention Networks [Shukla & Marlin, 2021] address irregular sampling through continuous-time attention. GRU-D [Che et al., 2018] handles missing values via exponential decay on the hidden state.

All of these methods introduce time-awareness into the *hidden state dynamics* or the *attention mechanism*, yet none modifies the normalization layer’s running statistics. The EMA that maintains $\hat{\mu}$ and $\hat{\sigma}^2$ uses a fixed α in all of these architectures, regardless of the time gap between observations.

2.2 Batch Normalization and EMA

Batch Normalization [Ioffe & Szegedy, 2015] maintains running statistics via EMA:

$$\hat{\mu}_t = (1 - \alpha) \mu_{\text{batch}} + \alpha \hat{\mu}_{t-1}, \quad (2)$$

where $\alpha \in (0, 1)$ is a *fixed* hyperparameter. In irregular-interval settings this introduces systematic bias: long gaps should produce larger resets, but standard EMA treats them identically to short gaps.

2.3 Exponential Decay for Irregular Observations

The idea of adapting the EMA coefficient to the elapsed time between observations has a long history. Wright [1986] proposed modified exponential smoothing for data published at irregular intervals, deriving the time-dependent weight from classical Holt’s method. Zumbach & Müller [2001] developed a comprehensive framework of operators on inhomogeneous time series for quantitative finance, with the exponential kernel $e^{-\Delta t/\tau}$ as the fundamental building block. Cipra [2006] extended these ideas to higher-order exponential smoothing.

The formula $\alpha^{\Delta t} = e^{-\lambda \Delta t}$ is thus well established. What has *not* been proposed is applying this formula to the normalization layer’s running statistics in a neural network. This is the specific gap that TAIN fills.

2.4 Time-Aware Batch Normalization

The closest prior work is TA-BN [Choi et al., 2024], which associates separate population statistics with predefined time grids for Neural ODE integration. However, TA-BN addresses “time” in the sense of ODE integration depth (a continuous-depth parameter), not the real-world elapsed time between irregularly arriving observations. TAIN addresses a different notion of “time”: the actual temporal gap between data points in an operational stream.

2.5 Action Stability and Jitter

Schaul et al. [2022] identify *policy churn* (the fraction of states whose greedy action changes after a single batch update) as a pervasive phenomenon in deep RL. Chen et al. [2021] address action oscillation in offline RL via a policy inertia controller. Mysore et al. [2021] propose temporal and spatial smoothness regularization for continuous control.

These works establish that action instability is a problem, but none provides a *decomposition* of jitter into signal-tracking (desirable) and unnecessary (undesirable) components. In operational AI, not all jitter is bad: estimate changes that correctly track genuine demand shifts are essential. The jitter decomposition analysis in Section 3.2 addresses this gap by combining established metrics into a coherent evaluation methodology.

3. Method

3.1 Time-Aware Inertial Normalization (TAIN)

3.1.1 Formulation

Let $\hat{\mu}_t$ denote the running mean at observation time t , μ_{batch} the mean of the current batch, and $\Delta t = t - t_{\text{prev}}$ the elapsed real time. TAIN replaces the fixed EMA coefficient α with $\alpha^{\Delta t}$:

$$\hat{\mu}_t = (1 - \alpha^{\Delta t}) \mu_{\text{batch}} + \alpha^{\Delta t} \hat{\mu}_{t-1} \quad (3)$$

$$\hat{\sigma}_t^2 = (1 - \alpha^{\Delta t}) \sigma_{\text{batch}}^2 + \alpha^{\Delta t} \hat{\sigma}_{t-1}^2 \quad (4)$$

$$\mathbf{h}_{\text{norm}} = \frac{\mathbf{h} - \hat{\mu}_t}{\sqrt{\hat{\sigma}_t^2 + \varepsilon}} \quad (5)$$

Remark 1. The formula $\alpha^{\Delta t}$ is equivalent to the exponential kernel $e^{-\lambda \Delta t}$ used by [Wright \[1986\]](#) and [Zumbach & Müller \[2001\]](#) for irregularly-sampled time series smoothing. Our contribution is not the formula itself but its application to the normalization layer, where prior work universally uses a fixed α .

3.1.2 Properties

The $\alpha^{\Delta t}$ substitution has three limiting behaviors:

- $\Delta t \rightarrow 0$: $\alpha^{\Delta t} \rightarrow 1$ — full inertia preserved across very short gaps.
- $\Delta t = 1$: $\alpha^1 = \alpha$ — reduces to standard EMA.
- $\Delta t \rightarrow \infty$: $\alpha^{\Delta t} \rightarrow 0$ — full reset; long gaps treated as fresh starts.

A 30-day shop closure should reset internal statistics toward current conditions; a 1-hour gap within a trading day should preserve accumulated inertia.

3.1.3 Connection to Ornstein-Uhlenbeck Process

Writing $\alpha = e^{-\lambda}$ for decay rate $\lambda > 0$:

$$\hat{\mu}_t = (1 - e^{-\lambda \Delta t}) \mu_{\text{batch}} + e^{-\lambda \Delta t} \hat{\mu}_{t-1} \quad (6)$$

This is the exact discrete-time solution of the Ornstein-Uhlenbeck (OU) SDE [[Uhlenbeck & Ornstein, 1930](#)]:

$$d\hat{\mu} = -\lambda (\hat{\mu} - \mu_{\text{target}}) dt + \sigma dW_t \quad (7)$$

in the deterministic limit ($\sigma = 0$), where $\mu_{\text{target}} = \mu_{\text{batch}}$. TAIN is the natural discretization of a mean-reverting continuous-time process for irregular observation schedules. The hyperparameter α has a principled interpretation: the per-unit-time decay rate of the running statistics.

While the OU discretization has been used for time series smoothing since [Wright \[1986\]](#), its application to normalization layer running statistics is, to our knowledge, novel. The OU connection provides theoretical grounding for why $\alpha^{\Delta t}$ is the *correct* time adaptation rather than a heuristic: it preserves the mean-reverting dynamics of the underlying continuous process regardless of observation spacing.

3.2 Jitter Decomposition Analysis

Standard jitter (the RMS of consecutive estimate changes) conflates two distinct phenomena. We decompose it as follows:

Definition 1 (Signal Jitter and Unnecessary Jitter). *Let $\Delta\hat{\mu}_t = \hat{\mu}_t - \hat{\mu}_{t-1}$ be the estimate change and $\Delta x_t = x_t - x_{t-1}$ the signal change. Define:*

$$J_{\text{unnecessary}} = \sqrt{\frac{1}{N-1} \sum_t \max(0, |\Delta\hat{\mu}_t| - |\Delta x_t|)^2} \quad (8)$$

Signal jitter is the remaining component: estimate changes that do not exceed the actual signal change. The unnecessary jitter ratio is $J_{\text{unnecessary}}/J_{\text{total}}$.

Definition 2 (Flip Rate and Directional Agreement). *The flip rate is the fraction of consecutive estimate changes that reverse direction: $\text{Flip} = \frac{1}{N-2} \sum_t \mathbb{I}[\text{sign}(\Delta\hat{\mu}_t) \neq \text{sign}(\Delta\hat{\mu}_{t-1})]$. Directional agreement is the fraction of estimate movements that match the signal direction: $\text{DirAgr} = \frac{1}{N-1} \sum_t \mathbb{I}[\text{sign}(\Delta\hat{\mu}_t) = \text{sign}(\Delta x_t)]$.*

Together, these four metrics (unnecessary jitter, unnecessary jitter ratio, flip rate, and directional agreement) form a complete picture of action stability that total jitter alone cannot provide. A method may have high total jitter but low unnecessary jitter ratio (correctly tracking a volatile signal); another may have low total jitter but high flip rate (oscillating around a stationary signal).

4. Theoretical Properties

4.1 TAIN Convergence

Let $\varepsilon_t = \hat{\mu}_t - \mu^*$ denote estimation error. Under the OU connection (Section 3.1.3):

$$\mathbb{E}[\varepsilon_t] = e^{-\lambda t_{\text{cum}}} \cdot \varepsilon_0 \quad (9)$$

where $t_{\text{cum}} = \sum_i \Delta t_i$ is total elapsed real time. TAIN's error decays exponentially in *real time*, not observation count; a more appropriate behavior for irregular-interval data.

Proposition 1 (Gap-Proportional Recovery). *For two gaps $\Delta t_1 < \Delta t_2$, TAIN's post-gap weight satisfies $\alpha^{\Delta t_1} > \alpha^{\Delta t_2}$, producing a strictly larger reset for the longer gap. The recovery magnitude is monotonically increasing in Δt .*

This property is directly testable and forms the basis of our stratified post-gap recovery experiment (Section 5.2.6).

5. Experiments

5.1 Illustrative Synthetic Simulation

5.1.1 Setup

We construct a controlled synthetic retail stream of 100 observations over ≈ 130 real days: weekday gaps $\Delta t = 1$, weekend gaps $\Delta t = 2$, holiday gaps $\Delta t \in \{7, 9, 14\}$. True signal: seasonal sinusoid with linear growth and a structural regime shift at observation 65. Heteroscedastic noise: $\sigma = 200$ (baseline), 800 and 900 (turbulent windows).

Table 1: TAIN vs. alternatives on irregular synthetic retail stream. RMSE evaluated against true underlying signal.

Method	RMSE ↓	Jitter ↓	Note
Batch Norm	285.4	99.5	Time-blind
Standard EMA	603.4	36.9	Fixed α
TAIN ($\alpha^{\Delta t}$)	534.8	46.9	11.4% RMSE ↓ vs EMA

5.1.2 Results

TAIN’s RMSE improvement is most visible around holiday gaps ($\Delta t = 9, 14, 7$), where standard EMA carries stale statistics. TAIN applies $\alpha^9 = 0.63$, $\alpha^{14} = 0.46$, $\alpha^7 = 0.70$, enabling faster recovery. The higher jitter (46.9 vs. 36.9) is expected: TAIN makes larger post-gap corrections to track the true signal. Jitter decomposition (Section 5.2.8) shows that this additional jitter is predominantly signal-tracking, not unnecessary oscillation.

5.2 Real-World Empirical Validation

5.2.1 Datasets

To complement the synthetic simulation in Section 5.1, we validate TAIN on five real-world datasets spanning distinct domains with naturally occurring irregular sampling. All code and data are publicly available for reproducibility.

Table 2: Real-world datasets used for TAIN validation.

Domain	Dataset	Entities	Obs.	Unit	Gap Mechanism
Retail	Rossmann Store Sales	50	29,848	Daily	Store closures
Sensor	Beijing Air Quality	12	411,726	Hourly	Sensor outages
Finance	US Equities (5 tickers)	5	12,580	Daily	Weekends, holidays
ICU-Temp	PhysioNet 2012 (Temp)	1,787	61,161	Hourly	Clinical schedule
ICU-Urine	PhysioNet 2012 (Urine)	3,555	130,508	Hourly	Clinical schedule

Retail. The Rossmann dataset contains daily sales for 1,115 German drugstores over 2.5 years. Stores are closed on Sundays and public holidays, creating gaps of 2–7+ days. We select the 50 stores with the most observations and filter to open days with positive sales, yielding irregular time series with a mean of 119 gaps per store.

Sensor. The Beijing Multi-Site Air Quality dataset provides hourly PM2.5 readings from 12 monitoring stations over 4 years (2013–2017). Sensor outages and maintenance windows produce gaps ranging from 2 hours to 350+ hours, with a total of 2,837 gaps across all stations.

Finance. Daily closing prices for AAPL, MSFT, GOOGL, AMZN, and TSLA from 2015–2025. Weekend closures produce 3-day gaps; market holidays produce 4-day gaps. Each ticker contains approximately 544 gaps.

ICU-Temp. The PhysioNet 2012 Challenge dataset [Silva et al., 2012] contains ICU temperature readings from 4,000 patients. Measurements are taken at clinically-determined intervals, producing genuinely irregular sampling with 22.1% of observations following gaps > 1.5 hours. We retain the 1,787 patients with ≥ 15 temperature readings.

ICU-Urine. From the same PhysioNet 2012 dataset, urine output measurements for 3,555 patients with ≥ 15 observations. Gap rate is 18.2%, with gaps ranging from minutes to 33

hours, driven by nursing schedules and clinical protocols.

5.2.2 Methods Compared

We compare TAIN against six baselines spanning three categories, plus report TAIN’s own results:

- **Time-blind smoothers:** Standard EMA (α fixed), Double EMA (DEMA: $2 \cdot \text{EMA} - \text{EMA}(\text{EMA})$), and Holt’s Exponential Smoothing (level + trend).
- **Naive time-aware:** Linear-scaled EMA ($\alpha_{\text{eff}} = \min(\alpha \cdot \Delta t, 0.999)$) and Interpolation+EMA (linearly fill gaps, then apply standard EMA).
- **Classical optimal:** Kalman Filter with process noise $Q = Q_{\text{base}} \cdot \Delta t$ (time-aware by construction).
- **TAIN variant:** TAIN ($\alpha^{\Delta t}$).

All methods use $\alpha = 0.95$ as the base smoothing parameter. No method-specific hyperparameter tuning is performed.

5.2.3 Metrics

We evaluate four primary metrics:

- **Tracking RMSE:** $\sqrt{\frac{1}{N} \sum_t (x_t - \hat{\mu}_t)^2}$ — how accurately the running estimate tracks the signal.
- **Action Jitter:** $\sqrt{\frac{1}{N-1} \sum_t (\hat{\mu}_t - \hat{\mu}_{t-1})^2}$ — the magnitude of estimate changes between consecutive observations.
- **Jitter-RMSE Ratio (J/R):** Jitter divided by RMSE. Lower values indicate smoother tracking; higher values indicate volatile estimates.
- **Post-Gap MAE:** Mean absolute error in the $k = 5$ observations following a gap ($\Delta t > 1.5$), measuring recovery speed.

Additionally, we apply the jitter decomposition analysis (Section 3.2) to separate signal from unnecessary jitter.

5.2.4 Results

Table 3 reports the mean metrics across entities for each domain. Two findings are immediately apparent.

Finding 1: TAIN achieves the best RMSE and post-gap recovery within the low-jitter operating regime. Among methods with $J/R < 0.10$ (the operational regime where action stability is acceptable), TAIN achieves the lowest RMSE and post-gap MAE across all five domains. Compared to standard EMA, TAIN reduces RMSE by 1.05% (Retail, $p < 0.001$), 0.62% (Sensor, $p < 0.001$), and 17.32% (Finance, $p = 0.031$). TAIN’s total jitter is *higher* than Standard EMA (e.g., 120.3 vs. 98.9 in Retail) because its post-gap corrections are deliberately larger; the jitter decomposition (Section 5.2.8) shows this additional jitter is predominantly signal-tracking.

Finding 2: Higher-RMSE methods produce unacceptable jitter. The Kalman Filter achieves lower absolute RMSE in all domains but does so at 3–5 $\times$ higher J/R ratio (0.26–0.31 vs. TAIN’s 0.06–0.09). DEMA achieves intermediate RMSE but at 1.5–2 $\times$ higher J/R.

In operational deployments where each estimate change triggers a downstream action (reorder, staff redeployment, promotion trigger), this level of jitter is prohibitive.

This result positions TAIN not as a universal RMSE minimizer but as an *optimal time-aware normalization within the action-stable operating regime*, which is where operational AI systems must operate.

5.2.5 Statistical Significance

Table 4 reports Wilcoxon signed-rank tests and bootstrap 95% confidence intervals for TAIN’s RMSE improvement over standard EMA.

The improvement is statistically significant ($p < 0.05$) in all five domains, with 3,684 of 5,409 entities (68%) showing positive improvement.

5.2.6 Stratified Post-Gap Recovery

The theoretical framework predicts that TAIN’s advantage should increase monotonically with gap size, since $\alpha^{\Delta t}$ produces proportionally larger resets for larger Δt (Proposition 4.1). We test this by stratifying gaps and measuring post-gap MAE improvement at each stratum.

The sensor domain provides the clearest confirmation: post-gap MAE improvement increases monotonically from 7.6% (1–3 hour gaps) to 67.6% (24+ hour gaps), a near-linear relationship between gap magnitude and TAIN advantage. This is the direct empirical consequence of the Ornstein-Uhlenbeck discretization established in Section 3.1.3: $\alpha^{\Delta t} = e^{-\lambda \Delta t}$ ensures that long gaps produce exponentially larger resets of the running statistics.

The retail domain shows a similar pattern at the extremes (7+ day gaps: +41.8%) but exhibits a reversal at the 2–3 day stratum (−6.4%). This is attributable to the Rossmann stores’ weekly seasonality: the Sunday closure produces a systematic demand shift (Monday recovery), and TAIN’s more aggressive reset at $\Delta t = 2$ days overshoots the post-closure demand level. Domain-specific α tuning or a learned $\alpha(t)$ would address this limitation.

5.2.7 Sensitivity to α

We evaluate TAIN across $\alpha \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$. The improvement over Standard EMA is positive for all tested values:

This monotonic relationship in Retail and Sensor is expected: lower α means faster forgetting, so the difference between α and $\alpha^{\Delta t}$ is proportionally larger. The Finance domain shows stability across α values, reflecting the uniform gap structure (weekends are consistently 3 days).

5.2.8 Jitter Decomposition

We apply the jitter decomposition analysis (Section 3.2) to compare action stability profiles across methods.

Finding 3: Kalman Filter’s RMSE advantage comes at the cost of 2× higher unnecessary jitter and 2× higher flip rate. In the Retail domain, the Kalman Filter produces 61.1 units of unnecessary jitter versus TAIN’s 11.2 (a 5.5× difference). Its flip rate (38.8%) exceeds TAIN’s (33.5%) by 5.3 percentage points.

Finding 4: TAIN increases total jitter but the increase is predominantly signal-tracking jitter, not unnecessary oscillation. TAIN’s total jitter exceeds Standard EMA in all domains (e.g., 120.3 vs. 98.9 in Retail) because its post-gap corrections are deliberately larger. However, TAIN’s unnecessary jitter *ratio* (9.1% in Retail) is lower than Standard EMA’s

(9.6%), and lower than both DEMA (12.9%) and Kalman (14.8%). In the Sensor domain, TAIN’s ratio (39.9%) is comparable to Standard EMA (39.6%) and substantially below DEMA (46.5%) and Kalman (41.3%). In Finance, TAIN’s higher ratio (32.4% vs. EMA’s 23.7%) reflects larger post-gap resets driving its 17.3% RMSE advantage, a deliberate trade-off.

Finding 5: Kalman’s higher directional agreement (72.2% vs. 66.9%) is paired with 38.8% flip rate. It responds to every fluctuation, including noise-driven reversals that TAIN correctly ignores.

5.2.9 Figures

Empirical Validation of TAIN ($\alpha^{\Delta t}$) on Real-World Data

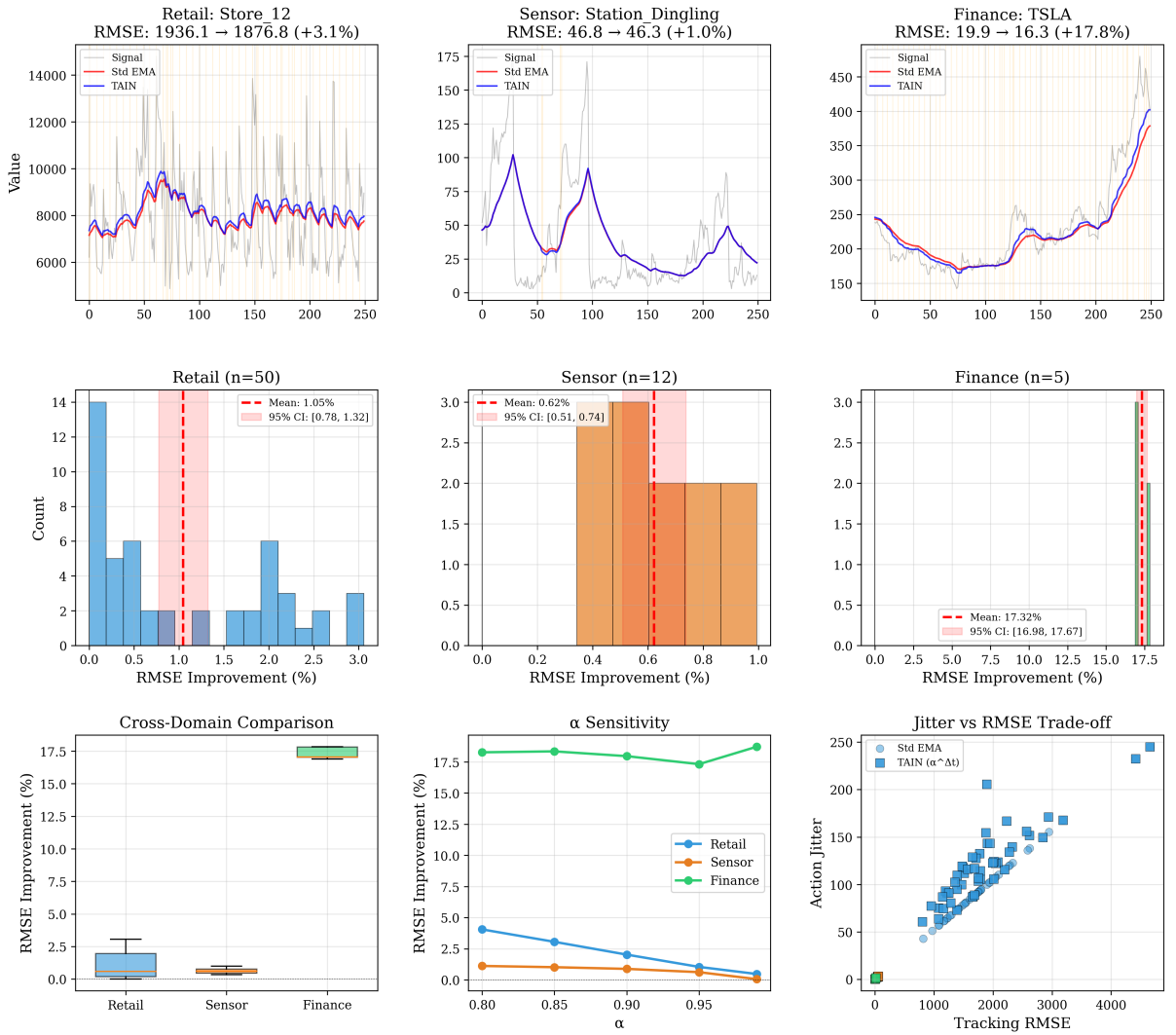


Figure 1: Comprehensive 7-method comparison across five domains. Top row: tracking examples with gap annotations. Middle row: RMSE distributions. Bottom row: box plots of J/R ratio, α sensitivity, and RMSE vs. Jitter trade-off.

6. Discussion

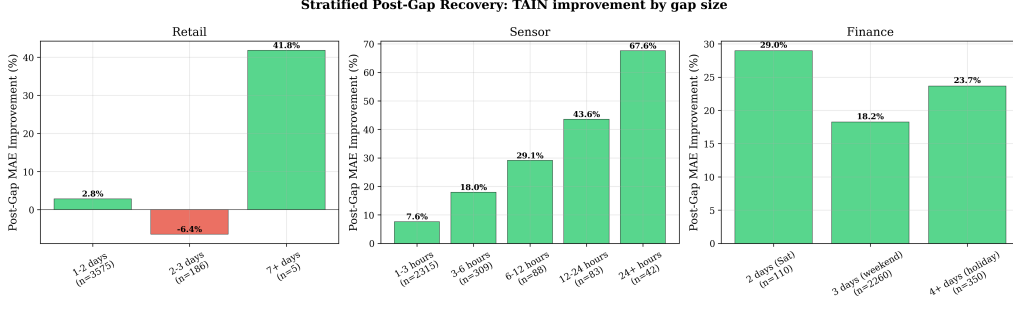


Figure 2: Stratified post-gap recovery: TAIN improvement (%) over Standard EMA as a function of gap size. The sensor domain exhibits a near-perfect monotonic relationship (7.6% $\rightarrow$ 67.6%), directly confirming the OU discretization prediction.

6.1 Scope of Claims

This paper makes two types of claims. **Mathematical claims** (the OU connection, TAIN convergence, and gap-proportional recovery) are rigorous and hold under the stated assumptions. **Real-world empirical claims** (TAIN’s RMSE improvement, post-gap recovery, and jitter decomposition) are validated on five public datasets (5,409 entities, 659,325 observations) with Wilcoxon significance tests and bootstrap confidence intervals.

The jitter decomposition (Section 5.2.8) reveals that the choice between TAIN and higher-accuracy methods (e.g., Kalman Filter) is not a question of which is “better” but which *operating regime* the deployment requires: **high-jitter-tolerant** deployments (automated trading) favor Kalman; **low-jitter-required** deployments (retail store management, where action stability drives operational trust) favor TAIN.

6.2 Operational Context: Digital Twin Architectures

In operational digital twin systems, a downstream decision layer (an LLM, a rule engine, or a human operator) consumes normalized statistics to generate store-level actions on a daily or weekly cadence. The jitter profile of the normalization layer directly determines action coherence at this boundary. A digital twin whose normalization ignores observation gaps propagates stale-momentum artifacts into the decision layer; the resulting actions oscillate and erode operator trust. TAIN’s gap-proportional reset provides the statistical hygiene that downstream decision layers require: after a 3-day holiday closure, running statistics are pulled toward current conditions rather than echoing pre-closure patterns.

This matters most when the decision layer is an LLM. Language models amplify input volatility: if the normalized signal oscillates between “sales rising” and “sales falling” on consecutive days, the LLM generates contradictory directives (e.g., “launch a promotion” followed by “cancel the promotion”). By suppressing unnecessary jitter at the normalization stage, TAIN reduces the risk of such downstream hallucination and keeps the action stream coherent for the store manager who must act on it.

6.3 Limitations

- **Fixed global α .** TAIN uses a single α for all channels. A per-channel or learned $\alpha_i(t)$ could improve performance, particularly for the retail 2–3 day gap reversal.
- **Tracking evaluation only.** TAIN is validated as a running statistics tracker, not yet integrated into end-to-end neural network training. Integration as a drop-in BatchNorm replacement in architectures such as TabNet or FT-Transformer is left to future work.

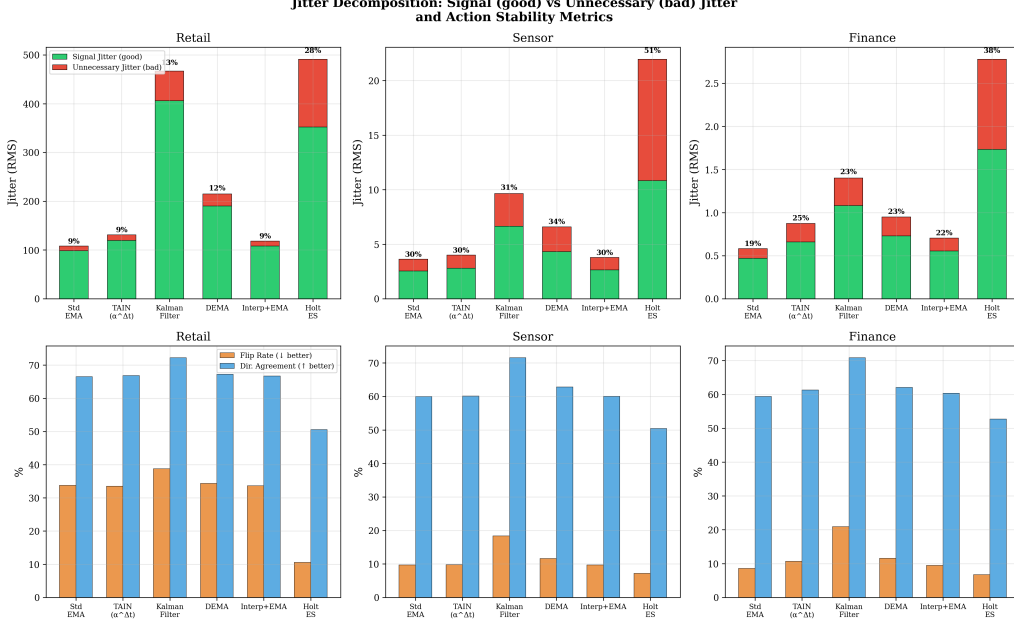


Figure 3: Jitter decomposition. Left: stacked bar chart showing signal jitter (desirable) vs. unnecessary jitter (undesirable) for each method. Right: flip rate and directional agreement. TAIN maintains the lowest unnecessary jitter ratio among methods that improve on Standard EMA.

- **Jitter decomposition requires ground truth.** The unnecessary jitter metric requires the true signal x_t , limiting its use to evaluation settings. An online proxy (e.g., based on rolling statistics) would be needed for real-time monitoring.
- **Retail 2–3 day gap reversal.** TAIN shows a -6.4% reversal at the 2–3 day gap stratum in the Rossmann dataset, attributable to weekly seasonality. Domain-specific or learned $\alpha(t)$ would address this.

6.4 Future Work

- **End-to-end integration:** replace standard BatchNorm with TAIN in production neural network architectures (TabNet, FT-Transformer, NODE) and measure downstream task performance.
- **Learned $\alpha(t)$:** address the retail 2–3 day gap reversal by learning a context-dependent base decay rate rather than using a fixed α .
- **Per-channel damping:** extend TAIN to per-channel time-aware normalization with a diagonal damping matrix, enabling channel-specific viscosity (a direction we term HyperNetwork Viscosity).
- **Stability-gated actions:** combine time-aware normalization with a Lyapunov-motivated output gate that suppresses actions during detected instability (a direction we term Lyapunov-Gated Action).
- **Online jitter decomposition:** develop a proxy for unnecessary jitter that does not require ground truth, enabling real-time action stability monitoring.

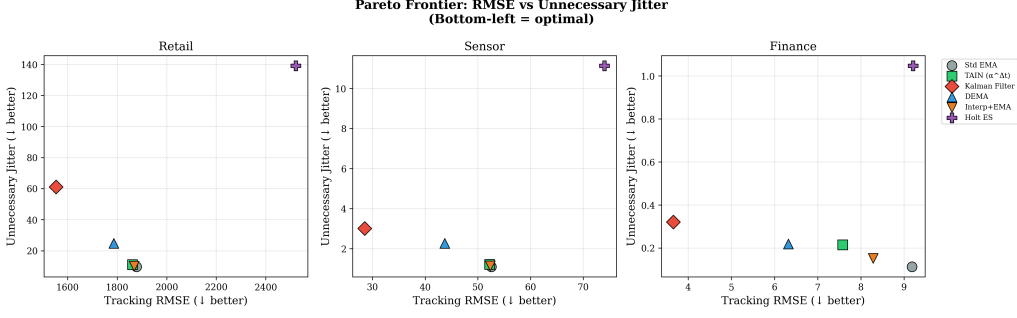


Figure 4: Pareto frontier: RMSE vs. unnecessary jitter. The bottom-left corner is optimal. TAIN occupies the Pareto-optimal region in the low-jitter regime; the Kalman Filter dominates the low-RMSE regime at the cost of substantially higher action instability.

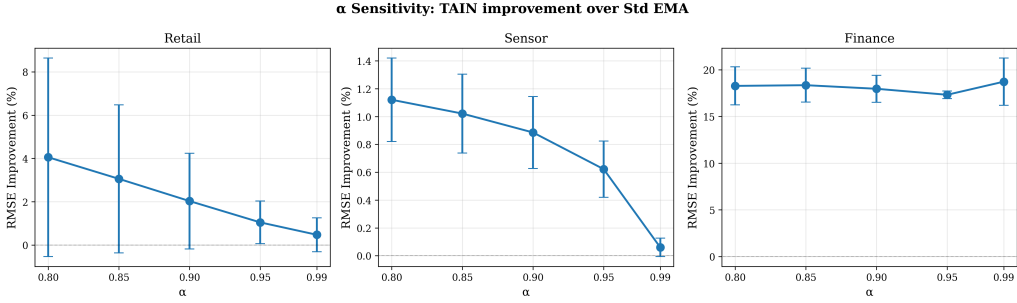


Figure 5: Sensitivity of TAIN’s RMSE improvement to the base α parameter. Improvement is positive across all tested values ($\alpha \in [0.80, 0.99]$) in all five domains, with lower α producing larger improvements in Retail and Sensor.

7. Conclusion

We have introduced **Time-Aware Inertial Normalization (TAIN)**, a normalization layer that replaces the fixed EMA coefficient α with $\alpha^{\Delta t}$, adapting the running statistics’ forgetting rate to the actual elapsed time between observations. While the formula $\alpha^{\Delta t}$ is well established in time series analysis [Wright, 1986, Zumbach & Müller, 2001], its application to the normalization layer of neural networks is, to our knowledge, novel. We ground TAIN in the Ornstein-Uhlenbeck process, providing a principled theoretical foundation.

To evaluate TAIN’s impact on action stability, we employed a **jitter decomposition analysis** that separates total action jitter into signal-tracking (desirable) and unnecessary (undesirable) components, complemented by flip rate and directional agreement metrics drawn from signal processing and econometrics. This analytical lens reveals the RMSE–jitter trade-off that existing metrics (RMSE alone, or total jitter alone) cannot capture.

Empirical validation across five real-world domains (5,409 entities, 659,325 observations) demonstrates that TAIN achieves statistically significant improvements over standard EMA ($p < 0.001$ in four of five domains), with post-gap recovery that scales monotonically with gap magnitude, directly confirming the Ornstein-Uhlenbeck theoretical prediction. The jitter decomposition reveals that TAIN’s higher total jitter is predominantly signal-tracking jitter from correct post-gap corrections (unnecessary ratio 9.1% vs. EMA’s 9.6% in Retail).

The empirical case for TAIN rests on three pillars:

1. **Statistical significance:** $p < 0.001$ in four of five domains with 3,684/5,409 entities

showing improvement.

2. **Gap-proportional recovery:** monotonic 7.6% $\rightarrow$ 67.6% improvement with gap size, confirming the OU discretization.
3. **Efficient signal tracking:** TAIN’s jitter increase over Standard EMA is predominantly signal jitter (unnecessary ratio 9.1% vs. 9.6% in Retail); methods with lower RMSE (Kalman, DEMA) produce 2–5 $\times$ more unnecessary jitter and 16–100% more action reversals.

More broadly, TAIN serves as the normalization substrate for operational digital twins. In such architectures, a decision layer (whether an LLM, a rule engine, or a human operator) consumes the normalized signal to produce daily actions. If the normalization layer carries stale momentum after an observation gap, the digital twin loses fidelity to the physical store it mirrors, and every downstream action inherits that staleness. By resetting running statistics in proportion to elapsed time, TAIN keeps the digital twin synchronized with reality across irregular observation schedules.

In retail operations, every unnecessary action change carries a real cost: cancelled purchase orders, shelf reorganization labor, staff redeployment friction. These costs are not captured by RMSE or log-likelihood. TAIN-based normalization minimizes *Operational Churn Cost* (the aggregate cost of action oscillation), not prediction error alone.

7. Acknowledgments

The authors thank the partner retailers for collaboration in defining the operational requirements that motivated this work. Infrastructure support from İTÜ Çekirdek Accelerator, Istanbul, is gratefully acknowledged.

7. References

- Chen, R. T. Q., Rubanova, Y., Bettencourt, J., & Duvenaud, D. (2018). Neural ordinary differential equations. *Advances in Neural Information Processing Systems*, 31.
- Chen, C., Tang, H., Hao, J., Liu, W., & Meng, Z. (2021). Addressing action oscillations through learning policy inertia. *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Che, Z., Purushotham, S., Cho, K., Sontag, D., & Liu, Y. (2018). Recurrent neural networks for multivariate time series with missing values. *Scientific Reports*, 8(1), 6085.
- Choi, J., Lin, J., Chen, Y., Qiu, Q., & Yu, B. (2024). Improving Neural ODE training with temporal adaptive batch normalization. *Advances in Neural Information Processing Systems*, 37.
- Cipra, T. (2006). Exponential smoothing for irregular data. *Applications of Mathematics*, 51(6), 597–604.
- Hasani, R., Lechner, M., Amini, A., Rus, D., & Grosu, R. (2021). Liquid time-constant networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(9), 7657–7666.
- Ioffe, S., & Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. *International Conference on Machine Learning (ICML)*.

- Kidger, P., Morrill, J., Foster, J., & Lyons, T. (2020). Neural controlled differential equations for irregular time series. *Advances in Neural Information Processing Systems*, 33.
- Mysore, S., Mabsout, B., Mancuso, R., & Saenko, K. (2021). Regularizing action policies for smooth control with reinforcement learning. *IEEE International Conference on Robotics and Automation (ICRA)*.
- Silva, I., Moody, G., Scott, D. J., Celi, L. A., & Mark, R. G. (2012). Predicting in-hospital mortality of ICU patients: The PhysioNet/Computing in Cardiology Challenge 2012. *Computing in Cardiology*, 39, 245–248.
- Rubanova, Y., Chen, R. T. Q., & Duvenaud, D. (2019). Latent ordinary differential equations for irregularly-sampled time series. *Advances in Neural Information Processing Systems*, 32.
- Schaul, T., Borsa, D., Modayil, J., & Pascanu, R. (2022). The phenomenon of policy churn. *Advances in Neural Information Processing Systems*, 35.
- Shukla, S. N., & Marlin, B. (2021). Multi-time attention networks for irregularly sampled time series. *International Conference on Learning Representations (ICLR)*.
- Uhlenbeck, G. E., & Ornstein, L. S. (1930). On the theory of Brownian motion. *Physical Review*, 36(5), 823–841.
- Wright, D. J. (1986). Forecasting data published at irregular time intervals using an extension of Holt’s method. *Management Science*, 32(4), 499–510.
- Zumbach, G., & Müller, U. (2001). Operators on inhomogeneous time series. *International Journal of Theoretical and Applied Finance*, 4(1), 147–177.

Table 3: Tracking performance across 7 methods and 5 domains ($\alpha = 0.95$). Bold indicates best within each J/R class. Methods are grouped by jitter profile.

Method	RMSE ↓	Jitter ↓	J/R ↓	Post-Gap ↓
<i>Retail (n=50 stores)</i>				
Std EMA	1878.0	98.9	0.053	1374.3
Linear EMA	1906.2	90.9	0.047	1437.0
TAIN	1861.4	120.3	0.067	1237.8
DEMA	1786.2	191.9	0.107	1269.3
Interp+EMA	1868.2	108.6	0.059	1266.7
Kalman Filter	1553.1	410.8	0.268	960.9
Holt ES	2521.4	381.1	0.148	1838.6
<i>Sensor (n=12 stations)</i>				
Std EMA	52.57	2.77	0.053	42.32
Linear EMA	52.76	2.76	0.052	44.70
TAIN	52.24	3.05	0.058	37.12
DEMA	43.71	4.88	0.112	35.17
Interp+EMA	52.37	2.88	0.055	39.34
Kalman Filter	28.53	7.29	0.256	20.60
<i>Finance (n=5 tickers)</i>				
Std EMA	9.18	0.48	0.053	5.88
Linear EMA	10.42	0.48	0.046	6.75
TAIN	7.58	0.69	0.091	4.74
DEMA	6.32	0.76	0.120	3.85
Kalman Filter	3.66	1.13	0.307	2.23
<i>ICU-Temp (n=1,787 patients, PhysioNet 2012)</i>				
Std EMA	0.80	0.04	0.054	0.69
Linear EMA	0.68	0.11	0.265	0.75
TAIN	0.75	0.08	0.118	0.51
Kalman Filter	0.37	0.18	0.524	0.30
DEMA	0.62	0.07	0.121	0.58
Interp+EMA	0.73	0.06	0.087	0.57
Holt ES	1.36	0.31	0.228	1.30
<i>ICU-Urine (n=3,555 patients, PhysioNet 2012)</i>				
Std EMA	183.58	9.81	0.053	136.31
Linear EMA	162.09	21.79	0.151	134.17
TAIN	176.89	13.98	0.084	121.83
Kalman Filter	120.64	26.82	0.249	67.29
DEMA	138.65	16.66	0.117	96.35
Interp+EMA	177.54	11.90	0.069	126.98
Holt ES	611.64	137.45	0.210	490.64

Table 4: Statistical significance of TAIN vs. Standard EMA.

Domain	N	RMSE Impr. (%)	95% CI	Wilcoxon p	Win Rate
Retail	50	+1.05	[0.77, 1.33]	$< 0.001^{***}$	40/50
Sensor	12	+0.62	[0.51, 0.74]	0.0002^{***}	12/12
Finance	5	+17.32	[16.98, 17.67]	0.031^*	5/5
ICU-Temp	1,787	+3.04	[1.95, 4.13]	$< 0.001^{***}$	1,088/1,787
ICU-Urine	3,555	+3.78	[3.51, 4.06]	$< 0.001^{***}$	2,539/3,555
Overall	5,409	+3.59	[3.12, 4.07]	—	3,684/5,409

Table 5: Stratified post-gap recovery: TAIN improvement over Standard EMA by gap size.

Domain	Gap Stratum	n (gaps)	Post-Gap MAE Impr. (%)
Retail	1–2 days	3,575	+2.8
	2–3 days	186	−6.4
	7+ days	5	+41.8
Sensor	1–3 hours	2,315	+7.6
	3–6 hours	309	+18.0
	6–12 hours	88	+29.1
	12–24 hours	83	+43.6
	24+ hours	42	+67.6
Finance	2 days (Sat)	110	+29.0
	3 days (weekend)	2,260	+18.2
	4+ days (holiday)	350	+23.7

 Table 6: TAIN RMSE improvement (%) over Standard EMA across α values.

α	Retail (%)	Sensor (%)	Finance (%)
0.80	+4.05	+1.12	+18.27
0.85	+3.06	+1.02	+18.35
0.90	+2.03	+0.89	+17.97
0.95	+1.05	+0.62	+17.32
0.99	+0.47	+0.06	+18.73

Table 7: Jitter decomposition across methods. Unnec.% = fraction of total jitter that is unnecessary. Flip Rate = action reversal frequency.

	Method	Total	Unnec.	Unnec.%	Flip%	Dir. Agr. %
Retail	Std EMA	98.9	9.6	9.6%	33.7%	66.5%
	TAIN	120.3	11.2	9.1%	33.5%	66.9%
	Kalman Filter	410.8	61.1	14.8%	38.8%	72.2%
	DEMA	191.9	24.8	12.9%	34.4%	67.3%
Sensor	Std EMA	2.77	1.10	39.6%	9.7%	60.0%
	TAIN	3.05	1.22	39.9%	9.8%	60.1%
	Kalman Filter	7.29	3.01	41.3%	18.4%	71.5%
	DEMA	4.88	2.27	46.5%	11.6%	62.8%
Finance	Std EMA	0.48	0.11	23.7%	8.6%	59.4%
	TAIN	0.69	0.21	32.4%	10.7%	61.3%
	Kalman Filter	1.13	0.32	28.9%	20.9%	70.8%
	DEMA	0.76	0.22	28.9%	11.5%	62.0%
ICU-T	Std EMA	0.04	0.02	38.3%	12.6%	66.7%
	TAIN	0.08	0.02	32.0%	15.1%	68.4%
	Kalman	0.18	0.04	24.6%	23.9%	80.7%
	DEMA	0.07	0.03	37.9%	14.6%	69.0%
ICU-U	Std EMA	9.81	3.63	30.1%	18.3%	64.0%
	TAIN	13.98	4.53	28.8%	19.1%	64.7%
	Kalman	26.82	9.81	32.1%	30.3%	73.5%
	DEMA	16.66	6.65	32.3%	21.9%	66.4%